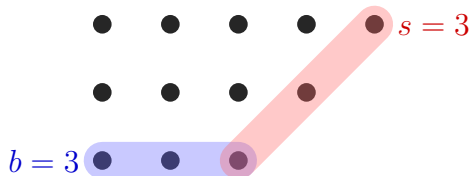


Exceptional Case 1: $b = s$

$$\{k, k+1, \dots, 2k-1\} \quad (k=3)$$

$$S = \{3, 4, 5\}$$



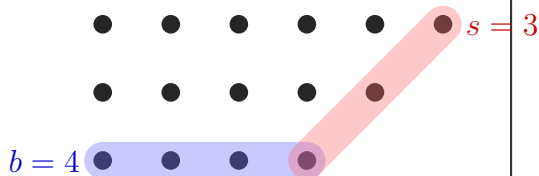
The Base and Slope intersect at the exact same element.
Removing b and moving s simultaneously is impossible.

$\phi(S)$ is **UNDEFINED**

Exceptional Case 2: $b = s + 1$

$$\{k+1, k+2, \dots, 2k\} \quad (k=3)$$

$$S = \{4, 5, 6\}$$



The Base and Slope intersect.
Moving the slope to the bottom row destroys the strict inequality of the distinct parts.

$\phi(S)$ is **UNDEFINED**

Trivial Case

The Empty Set

$$S = \emptyset$$



The empty set has no elements, so it has neither a base b nor a slope s .

$\phi(\emptyset)$ is **UNDEFINED**